What robot pretraining buys a vision-language-action model

A controlled component-interaction study across two embodiments

`Six arms · one frozen-backbone head · 2,600 closed-loop rollouts · LIBERO-Goal and ALOHA transfer-cube`

Existing work studies how to choose data and how to choose a backbone. This study asks what happens *between* the components: how backbone pretraining, camera configuration, action space and task structure interact to decide whether a policy actually succeeds in the loop.

> **Four findings.**
>
> 1. **Cameras dominate backbones.** Adding a wrist camera is worth +21 to +29 points; swapping a robot-pretrained backbone for its stock root is worth 2.5 points ($p = 0.40$).
> 2. **That null is bounded.** On a bimanual task the same backbone swap is worth **+9.5 points ($p = 0.0067$)**, and the entire gap is one transition — P(handover | lift), $p = 0.0009$.
> 3. **Pretraining buys ~2× training efficiency on the bimanual task** and the gap does not close by epoch 300 — but the same ladder on LIBERO shows no advantage at any checkpoint, so this is *not* a general sample-efficiency prior.
> 4. **Offline metrics cannot rank these policies.** On the bimanual pair they are not merely weak predictors; they carry no signal at all.

1. Design

One identical 19.2M-parameter head — token adapter plus DiT flow-matching decoder — is trained against **frozen** published backbones and compared by closed-loop rollout. Every arm shares the head, the read depth, the optimiser schedule and the evaluation protocol, so a difference in success rate is attributable to the factor that moved.

The pair spans one lineage: stock **Qwen3-VL-2B** → Cosmos-Reason2-2B → **GR00T N1.7**. Both hops are verified real finetunes (584/625 and 476/493 tensors differ), so the comparison covers the whole robot-pretraining treatment rather than one weak hop.

Two controls that make the comparison honest

Layer matching. Both arms are read at layer 16 — GR00T's own `select_layer`, intermediate for Qwen3-VL's 28 layers. Reading each at its own last layer would compare depth 16 against depth 28 and attribute a depth effect to pretraining.

Final-norm correction. Because layer 16 is final for one arm and intermediate for the other, the language stack's final RMSNorm is applied to the intermediate read. Without it the pair would differ by normalisation rather than by weights.

2. Cameras dominate backbones

LIBERO-Goal: 10 goals over one fixed scene, 200 rollouts per arm.

Arm	Backbone	Views	Val loss	Success	Swapped
exp03	GR00T N1.7	1	0.0316	62.5%	0.0%
exp04	Qwen3-VL-2B	1	0.0332	68.0%	0.0%
exp05	GR00T N1.7	2	0.0352	**91.5%**	0.0%
exp06	Qwen3-VL-2B	2	0.0352	**89.0%**	0.0%

The camera effect is **+29.0 points** (GR00T) and **+21.0 points** (Qwen3-VL), $p < 10^{-7}$. The backbone effect, measured at the benchmark's own two-camera specification, is **2.5 points** at $p = 0.40$. An order of magnitude separates the observation configuration from the backbone.

The instruction is load-bearing for every arm: **0/200** under a swapped instruction, all ten tasks. And both two-view arms converge to the same solution despite different backbone weights — wrist attention 51.6% vs 54.5%, ablation cost ×11.94 vs ×11.64.

3. The null does not survive a bimanual task

ALOHA transfer-cube was chosen as the pre-registered falsifier. Bimanual manipulation is inside GR00T's pretraining distribution, so this venue is biased *toward* finding a pretraining effect; a null here would have been strong evidence. It did not return a null.

Arm	Run 1	Run 2	Pooled (n=400)	Wilson 95% CI
GR00T N1.7	60.0%	62.5%	**61.25%** (245/400)	[56.4, 65.9]
Qwen3-VL-2B	49.0%	54.5%	51.75% (207/400)	[46.9, 56.6]

+9.5 points, z = 2.71, p = 0.0067 — clears the Bonferroni bar for the six comparisons in this study (0.05/6 = 0.0083), and the intervals do not overlap. The two runs used disjoint seed ranges, so this is a genuine replication rather than a resampling of the policy's own noise.

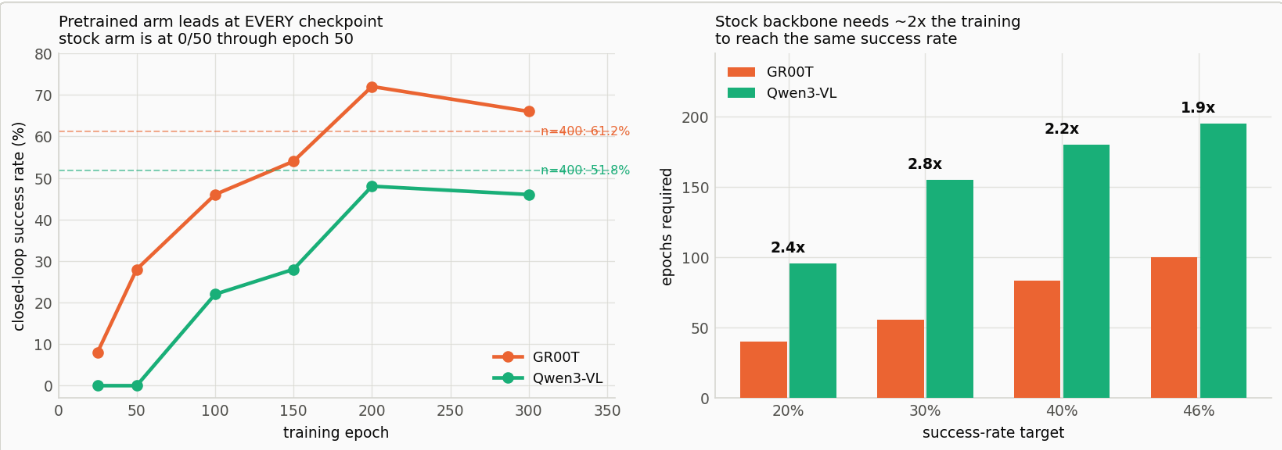
The gap is one transition, not general competence

ALOHA scores touch / lift / handover / success. `max_reward == 3` never occurs in 800 episodes, so once the receiving gripper contacts the cube the episode always completes, and the ladder is touch → lift → handover.

Stage	GR00T	Qwen3-VL	Δ	p
P(touch)	89.2%	92.5%	−3.3	0.11
P(lift ∣ touch)	95.8%	93.8%	+2.0	0.22
P(handover ∣ lift)	**71.6%**	**59.7%**	**+12.0**	**0.0009**

Both early stages run slightly *against* the pretrained arm. This is not a uniform competence advantage — it is localised to the one stage that bimanual pretraining plausibly covers, and the localisation is tighter than the top-line result.

4. Speed or skill?

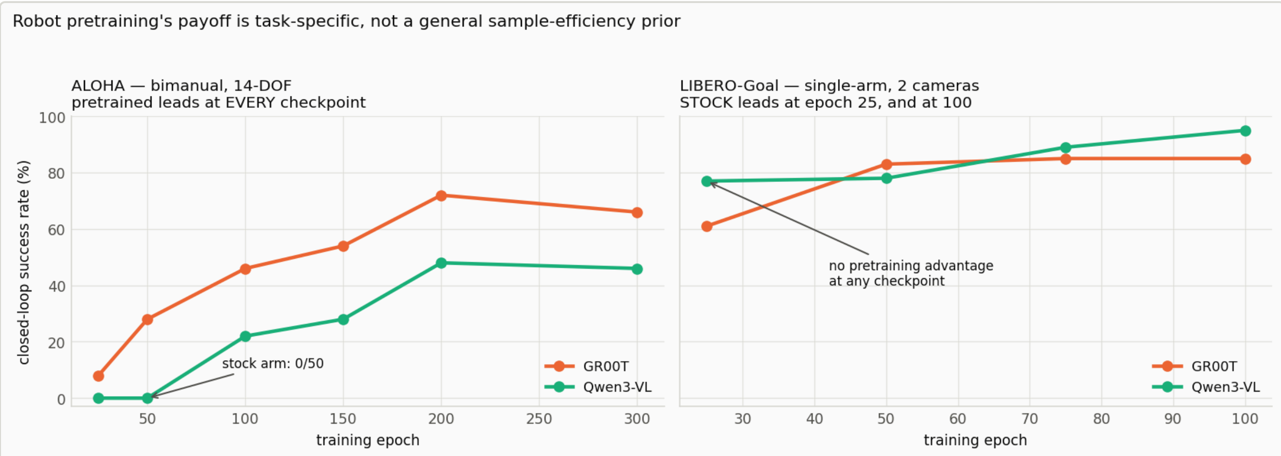


Closed-loop checkpoint ladder, paired seeds, 50 episodes per point. Left: the pretrained arm leads at every checkpoint and the stock arm scores 0/50 through epoch 50. Right: epochs required to reach a given success rate. Dashed lines mark the n=400 anchors — the ladder establishes the shape, the pooled runs the magnitude.

Epoch	25	50	100	150	200	300
GR00T	8.0%	28.0%	46.0%	54.0%	72.0%	66.0%
Qwen3-VL	**0.0%**	**0.0%**	22.0%	28.0%	48.0%	46.0%

- The stock backbone **cannot do the task early at all** — 0/50 at epochs 25 and 50 while the pretrained arm is at 28% (McNemar p = 0.0001).
- Stock needs **~2× the training** to reach any given rate: 2.4× / 2.8× / 2.2× / 1.9× at the 20/30/40/46% targets.
- The gap **does not close** by epoch 300.

The same ladder on LIBERO points the other way



Both checkpoint ladders on a common axis. Left: on bimanual ALOHA the pretrained arm leads at every checkpoint and the stock arm cannot do the task at all before epoch 100. Right: on single-arm LIBERO-Goal the stock arm leads at epoch 25 and again at epoch 100. The early-training effect does not transfer.

Epoch	25	50	75	100
GR00T	61.0%	83.0%	85.0%	85.0%
Qwen3-VL	**77.0%**	78.0%	89.0%	**95.0%**

n = 100 per point, canonical condition only. Pairing is exact and needs no seed offset — LIBERO supplies 50 *fixed* initial states per task, so every snapshot of every arm saw identical starts.

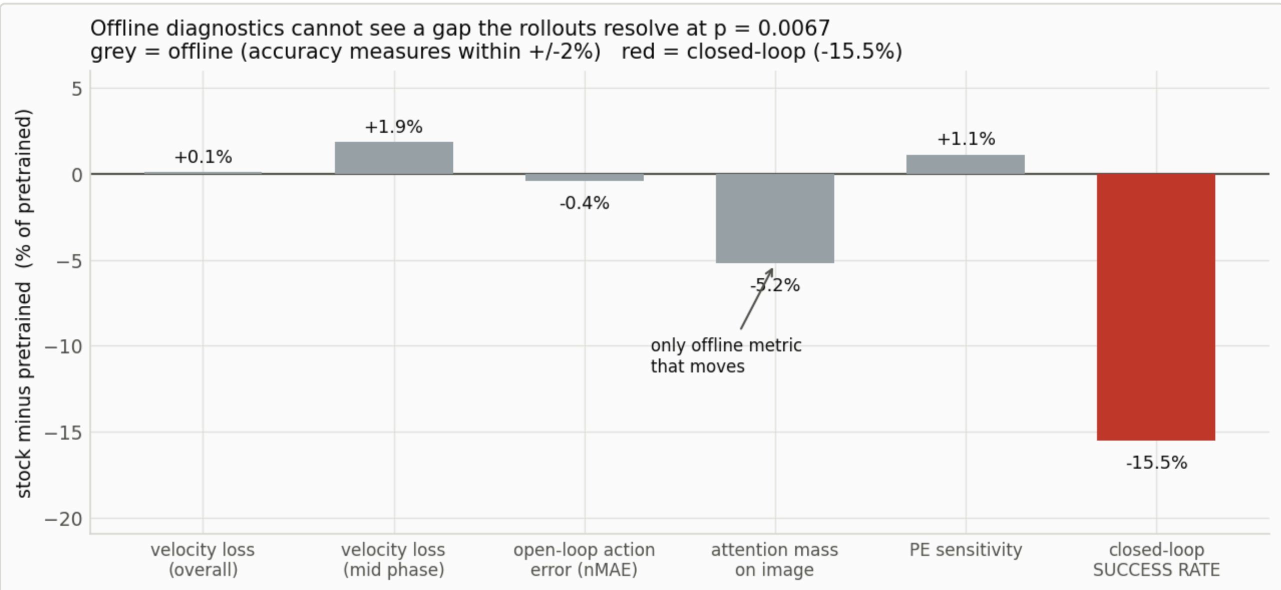
At epoch 25 the **stock** arm leads by 16 points, the opposite of ALOHA, where the pretrained arm led 8% to 0% at the same epoch and 28% to 0% by epoch 50. **ALOHA's early-training effect — the largest in this study — has no LIBERO counterpart.**

Neither significant point survives correction. Four comparisons, Bonferroni bar $0.05/4 = 0.0125$; the epoch-25 and epoch-100 gaps sit at $p = 0.0195$ and 0.0213. The defensible statement is the negative one: no LIBERO checkpoint shows a pretraining advantage, and the two points that reach nominal significance both favour the stock arm. A prediction was registered before this run that the LIBERO curves would *converge early*; they do not converge, they cross. The conclusion survives in a stronger form than predicted, but that specific prediction failed.

Why the ladder's magnitude is not quotable. Its 50 paired scenes are a fixed and slightly unrepresentative sample: at epoch 300 GR00T reads 66.0% against the 61.25% anchor (+4.8) and Qwen3-VL reads 46.0% against 51.75% (−5.8). Each deviation sits inside one standard error (~6.7 points), so nothing is broken — but the ladder establishes the *shape* and the pooled $n = 400$ runs establish the *magnitude* (+9.5 points). The +24-point ladder gap is not the population gap.

The LIBERO ladder carries the same warning and disagrees with its headline in *both* directions: at epoch 100 GR00T reads 85.0% against the 91.5% anchor and Qwen3-VL 95.0% against 89.0%. Two causes — `best.pt` is epoch 111 for both arms, past the ladder's last snapshot, and the ladder runs 10 episodes/task against the headline's 20, so it sees half the initial states.

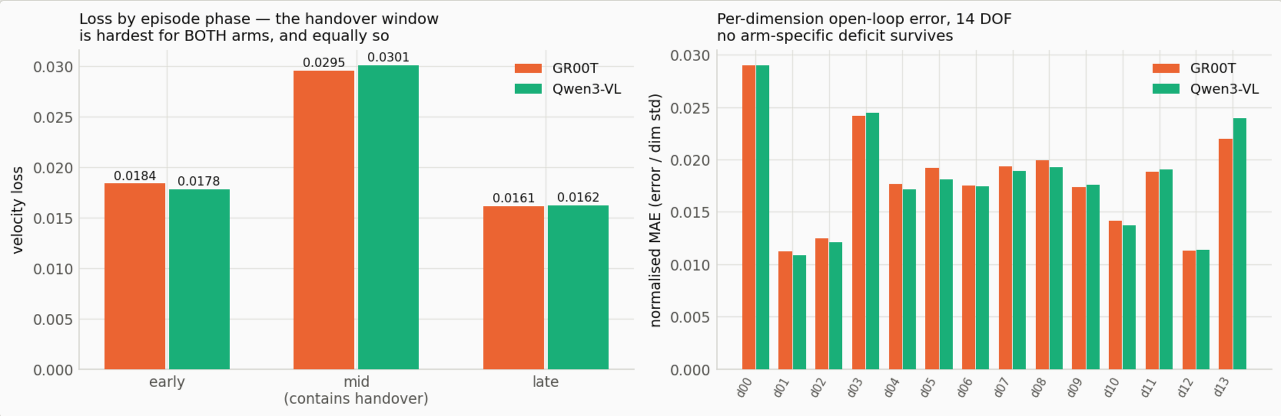
5. Offline metrics cannot rank these policies



Every offline diagnostic expressed as a between-arm difference, on the same axis as the closed-loop result. Four accuracy measures are flat against a gap resolved at $p = 0.0067$. Attention allocation is the only offline quantity that moves.

This is stronger than "offline ranks them wrongly." On the bimanual pair, offline carries **no signal** about a difference the rollouts resolve decisively. Loss by episode phase shows the objective *does* know the handover is hard — 0.0295 vs 0.0161 late — it simply does not know that one arm is worse at it.

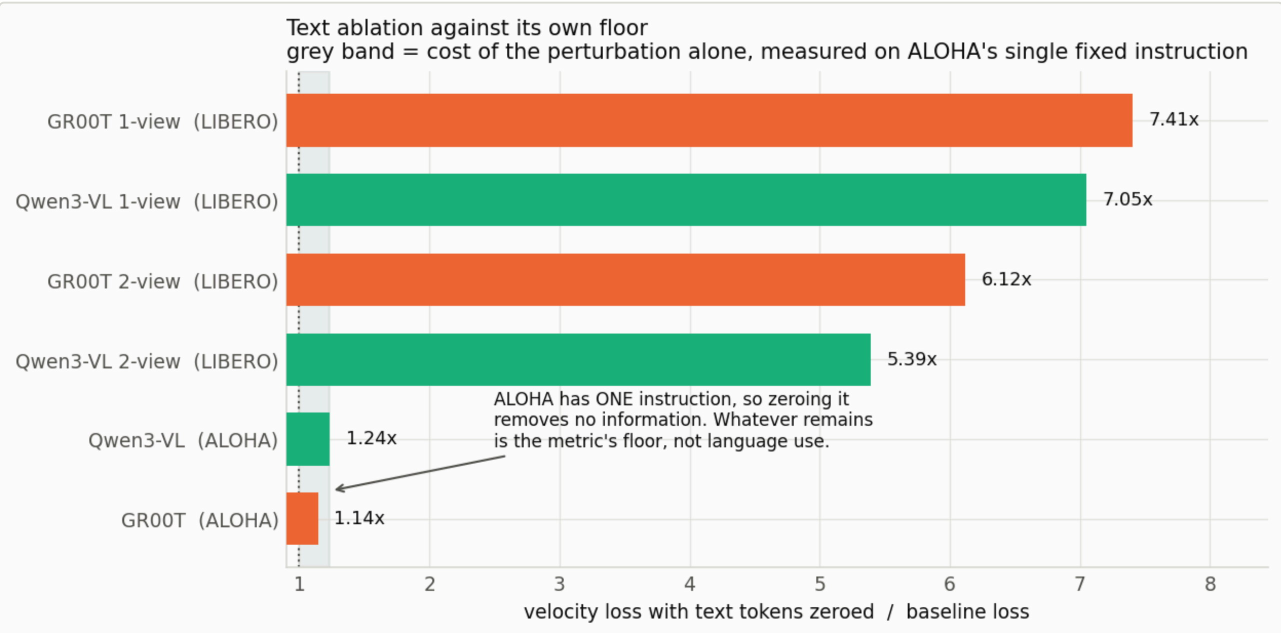
The one offline quantity that moves is attention *allocation*: the stock arm spends 5.2% less mass on image tokens and correspondingly more on text tokens that carry zero information on this task. Correlational, one testbed — the only surviving mechanistic candidate, not a finding.



Left: velocity loss by episode phase. The handover window is hardest for both arms and equally so. Right: per-dimension open-loop error across all 14 joints — no arm-specific deficit survives.

6. Two measurement lessons

Text ablation has a floor, and it must be measured

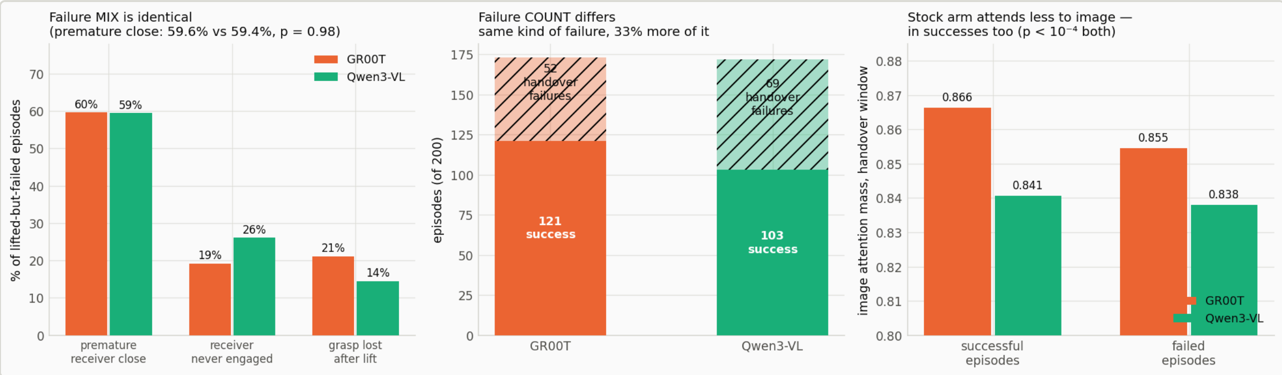


Text-ablation ratio across all six arms. ALOHA has a single fixed instruction, so its 1.14–1.24x is the cost of the perturbation alone — the floor against which the LIBERO ratios must be read.

Zeroing the text tokens of a *constant* instruction removes no task information, yet loss still rises 1.14×–1.24×. That residual is the cost of an off-distribution perturbation — the metric's floor, and a control condition an ablation on a multi-instruction dataset cannot supply for itself. Reporting an ablation ratio without its floor overstates the effect.

Note also that the two-view arms depend on text *less* than the one-view arms (5.39×/6.12× against 7.05×/7.41×): with a wrist camera available, some of what the instruction supplied becomes recoverable from vision.

What is physically different, if the action error is not?



Instrumented rollouts, n=200 per arm. Left: among episodes that lifted the cube but failed the handover, the failure MIX is statistically identical. Centre: only the count differs. Right: image-attention mass during the handover window, split by outcome so the comparison cannot be an artifact of failed episodes running longer.

The aggregate result records `max_reward`, which reports that an episode failed but not how. Instrumented rollouts log per-step 14-DOF state, commanded action, the reward timeline and per-replan attention. Arm identity was measured rather than assumed — the picking arm is whichever block moves first in successful episodes.

Failure class	**GR00T**	**Qwen3-VL**	**p**
premature receiver close	59.6% (31)	59.4% (41)	**0.98**
receiver never engaged	19.2% (10)	26.1% (18)	0.38
grasp lost after lift	21.2% (11)	14.5% (10)	0.34
total lifted-but-failed	**52**	**69**	—

> **The failure mix is identical; only the count differs.** The stock arm does not fail *differently* — it fails 33% more often in the same way. Premature closure of the receiving gripper is the dominant failure for both arms and this task.
>
> This answers why offline metrics miss the gap. If two policies fail the same way and differ only in how often they enter an unrecoverable configuration, there is no systematic per-step action-error signature to detect. The difference is a compounding closed-loop property, not a per-step accuracy property — exactly what an error averaged over a 16-step chunk and 2,000 frames cannot see.

Attention was recorded **during rollout**, not on dataset frames: dataset frames are successful expert demonstrations, so attention measured there cannot be linked to the policy's own failures. Over the five replans following the lift, the stock arm allocates less mass to image tokens — 0.8407 vs 0.8663 in successful episodes (Welch $t = 9.01$) and 0.8380 vs 0.8546 in failed ones ($t = 5.24$), both $p < 10^{-4}$. Because the difference is present in successes too, it is a stable policy-level difference in how the head uses its backbone, *not* a signature of impending failure.

The scene is not information-poor

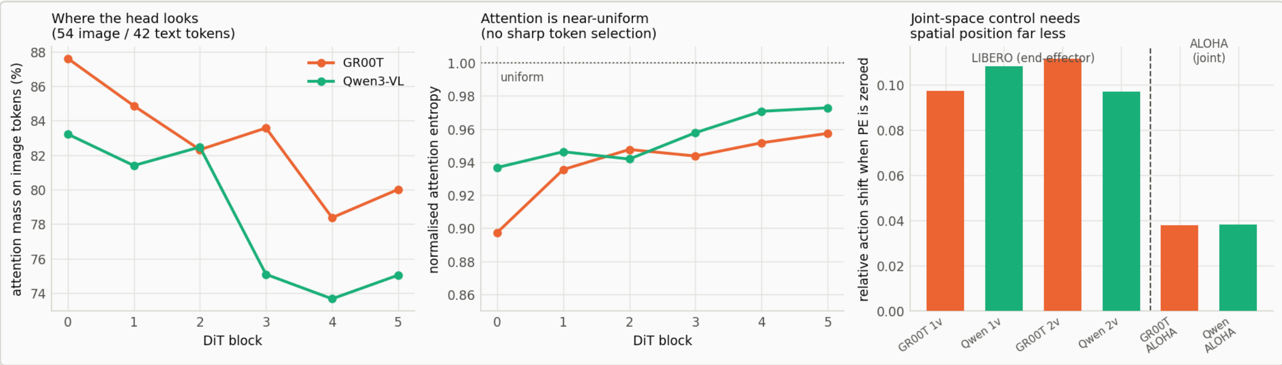
The positional-encoding result invites an objection: perhaps ALOHA's single fixed camera simply carries less spatial information. A gradient or Jacobian analysis cannot settle this — an impoverished scene produces low image sensitivity under either explanation. A cross-validated probe from image tokens alone to arm state can.

Arm	Image tokens	R^2(arm state)	R^2(action)
LIBERO GR00T 2-view	128	0.832	−0.215
LIBERO Qwen3-VL 2-view	128	0.828	−0.192
ALOHA GR00T	54	0.817	0.771
ALOHA Qwen3-VL	54	0.791	0.743

Arm configuration is **equally readable on both testbeds** (0.830 vs 0.804), and ALOHA reaches that from 54 image tokens against LIBERO's 128. The "visually simpler" objection has no support.

The second column sharpens the mechanism: action recoverability from one frame is −0.20 on LIBERO against +0.76 on ALOHA. LIBERO commands end-effector *deltas*, which a single frame does not determine; ALOHA commands *absolute joint targets*, which sit close to the pose the image already encodes. This is partly definitional, so it is reported as *explaining* the PE result rather than as independent evidence — what it establishes is that the operative variable is **what the action is defined relative to**, not degrees of freedom or arm count.

Action space determines how much the policy uses vision



Left and centre: cross-attention mass and entropy across the six DiT blocks. Right: PE sensitivity across all six arms — joint-space control needs image position far less than end-effector control does.

Zeroing the 2D positional encoding shifts actions by 0.097–0.111 under end-effector control but only **0.038** under joint control with full 14-DOF proprioception. Where the state already determines the arm configuration, the head sources far less from image position.

This was predicted the wrong way round before measurement — the pre-registered expectation was that bimanual handover would be *more* position-sensitive, on the reasoning that handover is spatial registration. It is not; the action space dominates.

7. Validity

The harness is correct. With the policy removed from the loop and the demonstrations' own actions replayed through the same env construction, success detector and step budget, LIBERO replays at 90.0% (no settling) and 92.0% (5 settling steps). That coincides with the band published methods report, so this is not a stricter harness. It is a replay rate, not a policy ceiling — the two-view arms exceed it on two tasks.

Training budget is not the constraint. Each arm takes 53,760 optimizer steps at batch 128 — 7.2× more samples and ~34× more passes over the data than a published LeRobot-family recipe on LIBERO. Validation flattens after epoch ~75 and open-loop action correlation reaches 0.979–0.997.

8. What this study cannot settle

The six-axis confound. LIBERO-Goal and ALOHA differ simultaneously in instruction variation (10 vs 1), degrees of freedom (7 vs 14), arm count, action space (end-effector vs joint), camera count (2 vs 1), and distribution match to GR00T's pretraining data. Two testbeds cannot separate six axes. The DOF hypothesis, the bimanual hypothesis and the distribution-match hypothesis all predict exactly what was observed, and this study cannot distinguish among them. Any statement naming one axis as *the* cause is an interpretation, not a result.

- **Two backbones, one lineage.** Every backbone claim rests on one pair.
- **One head architecture.** "Backbone barely matters on LIBERO" may be specific to a head with this capacity and cross-attention design.
- **Head diagnostics are correlational.** They locate where two heads differ on frozen checkpoints; they cannot prove that difference causes the handover gap. A causal test would swap handover-window tokens between arms.
- **Simulation only.** No physical robot.
- **Nulls are not equivalence.** At n = 200 per LIBERO condition the standard error is 3.3 points, so a null means "no detectable difference at this readout capacity," never "no difference."

9. Summary

Robot pretraining of a frozen VLM backbone is **not a general sample-efficiency prior**. Its payoff is task-specific, and on the testbed where it fails it fails at every point in training, not only at convergence.

On single-arm pick-and-place at the benchmark's own two-camera specification it contributes nothing measurable — 2.5 points at $p = 0.40$ at convergence, and no advantage at any checkpoint from epoch 25 onward — while a second camera contributes +21 to +29. On bimanual handover it makes the difference between 0% and 28% success at 50 epochs, cuts the epochs to any target by roughly half, and leaves a replicated +9.5-point advantage that localises to a single transition and does not close by epoch 300.

Throughout, offline velocity loss — the quantity these systems are trained on and selected by — failed to rank the resulting policies, and on the bimanual pair carried no signal about them at all.

Full design, statistics and defect log: `RESULTS.md` · Framing and component ledger: `OBJECTIVE.md`